Orbit and semidirect constructions: two complete originals

Human Compression project

Numerical candidate awaiting independent review

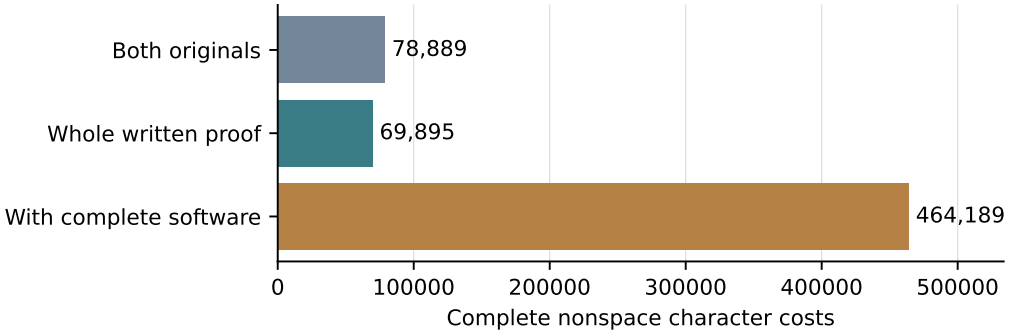
The complete written comparison is

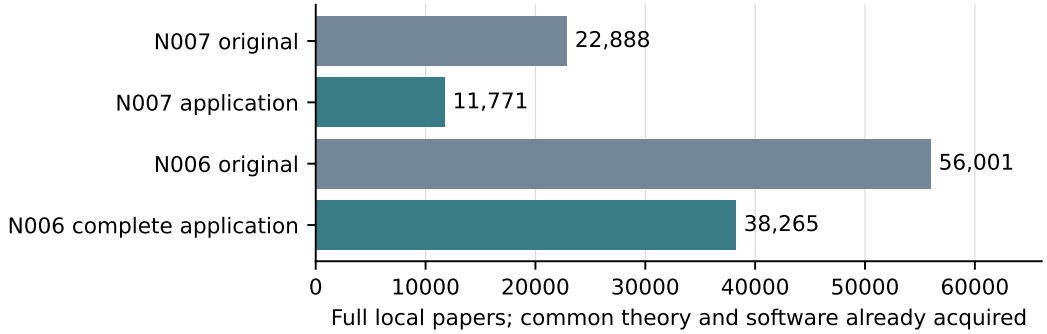
$$\frac{c(\mathrm{N007}) + c(\mathrm{N006})}{c(U) + c(A_{\mathrm{N007}}) + c(A_{\mathrm{N006}})} = \frac{78,889}{69,895} = \mathbf{1.128679}.$$

The whole eight-page common paper is charged once. The N007 application has five pages; the N006 application includes all three companion papers, of eight, five and four pages. Titles and references remain included. The prior N007-only edition is preserved separately.

Matched original	Original	Complete application	Conditional factor
N007: diagonal twistings	22,888	11,771	1.944440
N006: wreath and semidirect products	56,001	38,265	1.463505

The conditional figures assume that the *whole common theory and all declared software and required data have already been acquired*, including both decoding drivers. They do not omit any local companion paper. Common theory costs 19,859 characters.





Primary unit: calibrated, selected NFC nonspace characters. Source graphical substitutions are declared encoded-event proxies, not exact Unicode geometry. Raw and text-only sensitivities appear on the next page.

Complete software and the measurement limits

All nineteen whole GAP and witness files cost 383,279 characters; both complete required decoding drivers add 11,015. Including this entire 394,294-character software bundle gives

$$\frac{78,889}{69,895 + 394,294} = \frac{78,889}{464,189} = \mathbf{0.169950}.$$

Sixteen files provide finite proof inputs or complete saved witnesses, and three are validation files. Large saved permutations receive no discount. The four other operational drivers (20,299 characters) and three Lean files (5,820 characters, seventeen partial declarations) are reported separately.

There is no complete printed appendix for the eighteen new files. The earlier two-page appendix (3,161 characters) is an alternative for its single N007 validation file only. It cannot stand for all nineteen files, and it is never charged in addition to that same raw file.

NFC nonspace convention	Original total	Written total	Written factor	With software
Literal full extraction	79,300	69,925	1.134072	0.170825
Literal selected extraction	78,882	69,895	1.128579	0.169935
Calibrated selected (primary)	78,889	69,895	1.128679	0.169950
Identified proxies removed	78,449	69,895	1.122384	0.169002

All thirty ordinary coordinates (three extraction conventions, two normalizations and five units) are preserved. The ten additional text-only coordinates replace only identified graphical-proxy events by separators. That sensitivity does not measure the entire nontextual geometry. All forty settings and 160 exact fractions appear in the JSON and CSV records.

Separate external storage inventory	Bytes
Four required representation files	21,960,689
Sixteen listed installed data/algorithm files	2,516,249

These byte totals are separate from character costs. Binary/decoded and compressed/decompressed versions are alternative representations, not extra charges. The complete GAP runtime is trusted infrastructure and is not represented by the listed files. Retained classification proofs, historical software, full graphics and full formalization remain unknown, not zero.

This packet reuses the independently accepted input boundary and does not rerun source extraction, GAP or Lean. It covers the thirty-two N007 and twenty-seven corrected N006 clauses in the bound scientific edition. Later Fischer exception certificates lie outside this comparison. No source, question, priority or formal credit changes.